

## LESSON 3-2 • SESSION 1 • PRACTICE • FORM A

## Understanding Unit Rates

Learning target – I can find a unit rate and use rate language such as “per hour” or “each.” • 6.RP.A.2

**REMEMBER**

- **Per** means “for each.” A **unit rate** is the amount for **1** unit.
- Divide the first amount by the second: 90 pages  $\div$  6 minutes = 15 pages per minute.
- To predict, **multiply** the unit rate. To find a time or amount of units, **divide**.

**WORKED EXAMPLE**

A printer prints 90 pages in 6 minutes at a steady rate. How many pages in 7 minutes? How long for 45 pages?

**Step 1** Divide:  $90 \div 6 = 15$  pages per minute.

**Step 2** 7 minutes:  $15 \times 7 = 105$  pages.

**Step 3** 45 pages:  $45 \div 15 = 3$  minutes.

**Answer 105 pages; 3 minutes**

**WORDS** **rate** (*tasa*) • **unit rate** (*tasa unitaria*) • **per** (*por*) • **constant** (*constante*)

**PRACTICE**

show your work • label units

1. Find each unit rate.

a. 84 words in 7 min = \_\_\_\_ per min

b. \$56 for 8 hours = \_\_\_\_ per hour

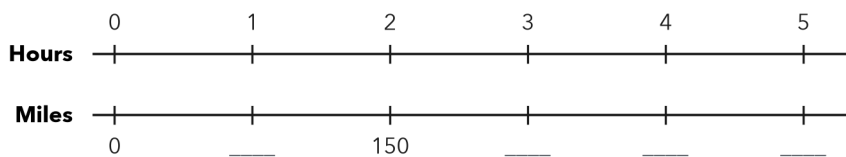
c. 108 miles on 4 gal = \_\_\_\_ per gal

d. 132 steps in 12 s = \_\_\_\_ per s

e. 96 beads on 6 bracelets = \_\_\_\_ each

f. 138 lb of soil in 6 beds = \_\_\_\_ per bed

2. **City Train.** A commuter train travels 150 miles in 2 hours at a steady speed. Complete the double number line.



a. Unit rate: \_\_\_\_\_ mph

b. Miles in 5 hours: \_\_\_\_\_ miles

3. **Lawn Mowing.** David earns \$45 for mowing 3 lawns. What is his rate per lawn?

**Hint** Divide dollars by lawns.

Show your work.

**Answer** \_\_\_\_\_ per lawn

4. **Juice Cost.** A 6-pack of juice costs \$7.50. Find the cost per bottle.

Show your work.

**Answer** \_\_\_\_\_ per bottle

5. **Gas Mileage.** A car travels 240 miles on 8 gallons of gas.

**Hint** For part b, divide the miles by the miles per gallon.

- a. Miles per gallon: \_\_\_\_\_ mpg  
b. Gallons needed for 360 miles: \_\_\_\_\_ gal

Show your work.

6. **Running Practice.** Carlos runs 12 miles in 2 hours.

**Hint** Time = distance ÷ speed.

- a. Speed: \_\_\_\_\_ mph  
b. Time to run 18 miles at this pace: \_\_\_\_\_ hours

Show your work.

7. **Two Jobs.** Job A pays \$54 for 6 hours. Job B pays \$85 for 10 hours.

- a. Hourly rate: Job A \_\_\_\_ Job B \_\_\_\_  
b. Which pays more per hour, and by how much? \_\_\_\_\_  
c. Total pay for 14 hours of Job B: \_\_\_\_\_ dollars

8. **Printer Rate.** A printer prints 288 pages in 8 minutes at a steady rate.

**Hint** Find pages for 1 minute, then multiply by 15.

- a. Pages per minute: \_\_\_\_\_  
b. Pages in 15 minutes: \_\_\_\_\_

## FIND THE ERROR

explain the mistake, then fix it

9. **Liam's Work.** A bike goes 45 miles in 3 hours. Liam calculated  $45 \times 3 = 135$  miles per hour. Explain his mistake and find the correct rate.

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## TEST PRACTICE

choose the best answer

10. A car travels 180 miles in 3 hours. The unit rate is –

- A 540 miles per hour      B 60 miles per hour  
C 183 miles per hour      D  $1/60$  mile per hour

11. 8 apples cost \$4.00. The unit price is –

- A \$0.50 per apple      B \$2.00 per apple      C \$32 per apple      D \$4.00 per apple

## SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

|         |             |           |         |        |         |         |          |         |          |
|---------|-------------|-----------|---------|--------|---------|---------|----------|---------|----------|
| 27<br>D | \$1.25<br>I | \$15<br>V | 24<br>X | 6<br>E | 75<br>I | 30<br>D | 300<br>Q | 36<br>S | 135<br>Z |
| #1c     |             | #2a       | #3      | #4     | #5a     | #6a     | #8a      |         |          |

## CHECK YOURSELF

- ☐ I can do this on my own    ☐ I need a little help    ☐ I have a question about #\_\_\_\_\_